

VELO

Athletic Intelligence

Project Design Document

Somnia Agentathon 2026 | Encode Club

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Hackathon	Somnia Agentathon — Encode Club
Track	Build the most novel and high-impact agent-driven application on Somnia
Duration	4 Weeks — May 20 to June 11, 2026
Platform	Web Application
Version	1.0 — Initial Design Document

1. Executive Summary

Velo is a multi-agent tennis coaching platform that autonomously analyzes player form, tracks performance trends over time, and generates verifiable training prescriptions — all logged on Somnia's Agentic Layer 1 infrastructure.

The platform is designed for tennis coaches who need to prepare athletes for specific opponents and evaluate training progress without spending hours reviewing footage manually. Velo reduces that process from hours to minutes by deploying AI agents that watch, reason, and prescribe — autonomously.

The core innovation is not the analysis itself. It is the verifiability. Every coaching prescription Velo generates is signed by an autonomous agent and written immutably on-chain, creating a tamper-proof record of what was recommended, when, and why. No other tennis coaching product does this.

2. The Problem

2.1 The Gap in Tennis Coaching Today

Tennis coaching at the competitive level is heavily dependent on video analysis. Coaches spend significant time reviewing match footage, noting opponent tendencies, identifying form breakdowns in their athletes, and building game plans. This process is almost entirely manual, time-consuming, and inconsistent across coaching teams.

At the professional level, teams have dedicated analysts and expensive software to support this work. At the development and semi-professional level — where the majority of competitive tennis happens — coaches do this alone, often with nothing more than a laptop and a video player.

2.2 Specific Pain Points

- Reviewing a single match for opponent tendencies can take two to four hours of manual work.
- Form analysis requires frame-by-frame scrubbing to identify biomechanical issues, which is tedious without specialist tools.
- Coaching decisions and prescribed drills are verbal or written in notebooks — there is no verifiable record of what was recommended or when.
- Existing AI sports tools are expensive, designed for professional teams, and not accessible to independent coaches.
- There is no neutral, tamper-proof record of training protocols — which matters for athlete development audits, federation compliance, and sponsorship accountability.

2.3 Why Existing Solutions Fall Short

Solution	Limitation
Dartfish / Hudl	Professional-tier pricing, not accessible to independent coaches
Generic AI video tools	No sport-specific intelligence, no coaching output
Manual video review	Time-intensive, inconsistent, no audit trail
Basic pose apps	Detection only, no reasoning, no prescriptions
Blockchain logging tools	No AI layer, no coaching intelligence

3. The Solution — Velo

3.1 What Velo Does

Velo accepts pre-recorded match or training footage via a web interface. Two autonomous AI agents process the footage, reason about what they observe, and write their outputs as verifiable records on Somnia's Layer 1.

A coach uploads a video, identifies the player, and within minutes receives a biomechanical form report and a structured training prescription — both generated autonomously by agents, both logged on-chain with a timestamp and agent signature.

3.2 The Unique Value Proposition

What Makes Velo Different

- Velo is the first platform where the agent is the coach — not a dashboard that reports what happened, but an autonomous system that prescribes what to do next.
- Every prescription is cryptographically signed and immutably logged on Somnia L1, creating verifiable coaching records that can be audited by federations, sponsors, and athlete managers.
- The system is accessible to any coach with a browser — no installation, no expensive hardware, no specialist data team required.
- The on-chain architecture means an athlete's performance history is portable, tamper-proof, and persistent across coaching relationships.

3.3 Target Users

User	Use Case
Development-level tennis coaches	Analyze training footage, generate drill prescriptions, track athlete progress
Academy coaching teams	Review opponent match footage before tournaments, build scouting reports
Athlete managers	Access verifiable on-chain performance history for sponsorship and federation reporting
Semi-professional players	Self-review form without a full coaching team

4. System Design

4.1 Architecture Overview

Velo is built on four layers that operate in sequence: a web frontend for coach interaction, a vision processing backend, an agent orchestration layer, and a Somnia on-chain layer for verifiable logging.

Layer	Components	Role
Frontend	React, Vite, TailwindCSS, ethers.js	Coach interface, video upload, wallet connection, results dashboard
Vision Backend	FastAPI, MediaPipe Pose, OpenCV, NumPy	Frame extraction, landmark detection, joint angle computation
Agent Layer	CrewAI, Marlin-2B VLM via HuggingFace API	Stroke reasoning, form evaluation, prescription generation
On-Chain Layer	Somnia L1, Solidity, Hardhat, MetaMask	Agent registry, orchestration contract, verifiable receipt logging
Storage	Supabase, Cloudinary	Player profiles, session metadata, video file storage
Hosting	Replit Pro	Full-stack deployment, public demo URL

4.2 Video Processing Pipeline

When a coach uploads footage, the FastAPI backend extracts frames at a defined interval using OpenCV. Each frame is passed to MediaPipe Pose, which returns 33 body landmarks with x, y, and z coordinates. NumPy computes joint angles at the shoulder, elbow, wrist, hip, and knee for each frame. The resulting angle data is structured as a JSON payload representing the player's biomechanical profile across the full video.

4.3 The Two Agents

Agent 1 — Form Analysis Agent

The Form Analysis Agent receives the structured landmark and angle data from the vision pipeline. It uses Marlin-2B, a compact Vision Language Model, to reason holistically about the player's stroke mechanics. Rather than applying simple threshold rules, Marlin-2B interprets the angle data in context — identifying compensatory movements, asymmetries, and phase-specific breakdowns across the preparation, contact, and follow-through phases of each stroke.

The agent outputs a structured form report identifying stroke-specific strengths and deviations, ranked by severity and linked to specific timestamps in the footage.

Agent 2 — Prescription Agent

The Prescription Agent receives the Form Analysis Agent's output and generates a training prescription — specific drills, focus areas, and session recommendations tailored to the identified form issues. This prescription is written in natural language, structured for direct use by a coach.

Critically, the Prescription Agent then calls the Somnia orchestrator smart contract, passing the prescription payload. The contract logs a verifiable on-chain receipt containing the player identifier, session timestamp, agent address, prescription hash, and a summary of findings. This receipt is permanent, tamper-proof, and queryable by the coach dashboard.

4.4 Somnia On-Chain Architecture

Two smart contracts are deployed on Somnia's Layer 1:

- Agent Registry Contract — registers Velo's agents on Somnia's agent discovery layer, making them callable and discoverable by other on-chain actors. This satisfies Somnia's agent-native composability requirement.
- Orchestrator Contract — receives structured analysis data from the FastAPI backend, routes it to the Form Analysis Agent, receives the Prescription Agent's output, and writes the final verifiable receipt on-chain. The contract emits an event on every successful prescription write, which the frontend listens to in real time.

The architecture ensures that no human action is required between video upload and on-chain receipt. The smart contract drives the agent invocation autonomously once triggered by the initial data submission.

4.5 Data Flow — End to End

Step	Action	Actor
1	Coach uploads video and identifies player via web interface	Coach (Frontend)
2	Frontend uploads video to Cloudinary, stores metadata in Supabase	Frontend + Supabase
3	FastAPI extracts frames, runs MediaPipe, computes joint angles	Vision Backend
4	Angle JSON payload sent to Orchestrator Contract on Somnia	FastAPI
5	Orchestrator Contract calls Form Analysis Agent	Somnia Smart Contract

6	Form Analysis Agent reasons over data via Marlin-2B, returns form report	Agent 1 (CrewAI + Marlin-2B)
7	Prescription Agent generates training prescription from form report	Agent 2 (CrewAI)
8	Orchestrator Contract writes verifiable receipt on-chain	Somnia Smart Contract
9	Frontend reads receipt, displays report and prescription to coach	Coach (Frontend)

5. Full Technology Stack

Category	Tool	Purpose
Vision	MediaPipe Pose	Player landmark detection and joint tracking
Vision	OpenCV	Frame extraction and video processing
Vision	NumPy	Joint angle computation from landmarks
AI	Marlin-2B VLM	Holistic biomechanical reasoning over frame and angle data
Agents	CrewAI	Agent orchestration, role definition, tool wiring
Backend	FastAPI (Python 3.11)	Vision pipeline API, WebSocket support, agent trigger
Frontend	React + Vite	Coach dashboard and interaction layer
Frontend	TailwindCSS	Rapid UI styling
Frontend	ethers.js	Somnia wallet connection and contract interaction
Blockchain	Somnia L1	Agent registry, orchestration, verifiable receipts
Blockchain	Solidity	Smart contract logic
Blockchain	Hardhat	Contract compilation, testing, deployment
Wallet	MetaMask	Coach authentication and on-chain signing
Database	Supabase	Player profiles, session metadata, history
Storage	Cloudinary	Video upload and storage
Hosting	Replit Pro	Full-stack deployment and demo URL
VLM Inference	HuggingFace Inference API	Marlin-2B inference without local GPU requirement
Version Control	GitHub	Repository (required per judging criteria)

6. Project Milestones

Week 1 — May 20 to May 27: Vision Pipeline and Project Foundation

The first week establishes the core vision processing pipeline and project infrastructure. This is the technical foundation everything else builds on.

Milestone	Deliverable	Status
Project scaffolding	GitHub repo, Replit environment, FastAPI skeleton, React shell	To Do
MediaPipe integration	Pose landmark extraction running on uploaded video	To Do
Angle computation	NumPy joint angle pipeline for shoulder, elbow, wrist, hip, knee	To Do
Stroke classification	Rule-based stroke phase detection from angle thresholds	To Do
Video upload	Cloudinary integration, Supabase session storage	To Do
Basic API	FastAPI endpoint accepting video, returning structured angle JSON	To Do
Attend workshop	Somnia How to Build session (May 22) — Somnia agent architecture	Scheduled

Week 2 — May 27 to June 3: Agent Orchestration

Week two builds the intelligence layer. CrewAI agents are defined, Marlin-2B is integrated, and the two agents are wired to process vision pipeline output end to end.

Milestone	Deliverable	Status
CrewAI setup	Agent definitions, roles, goals, and tool bindings for both agents	To Do
Marlin-2B integration	HuggingFace API call wired into Form Analysis Agent	To Do
Form Analysis Agent	Agent reasoning over angle data, producing structured form report	To Do
Prescription Agent	Agent generating natural language training prescription from form report	To Do
Agent pipeline test	End-to-end test: video in, prescription out, no on-chain yet	To Do

Dashboard v1	React frontend displaying form report and prescription output	To Do
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Week 3 — June 3 to June 8: Somnia On-Chain Integration

Week three wires the Somnia layer. Smart contracts are deployed, agents are registered, and the orchestrator contract drives the full autonomous pipeline.

Milestone	Deliverable	Status
Smart contracts	Agent Registry and Orchestrator contracts written in Solidity	To Do
Hardhat deployment	Contracts deployed to Somnia testnet	To Do
Agent registration	Velo agents registered on Somnia's agent discovery layer	To Do
Contract integration	FastAPI submits angle data to Orchestrator Contract	To Do
On-chain receipt	Prescription Agent writes verifiable receipt to Somnia L1	To Do
Frontend wallet	ethers.js + MetaMask connection, receipt display on dashboard	To Do
Autonomous flow test	Full pipeline test: upload to on-chain receipt, no manual steps	To Do

Week 4 — June 8 to June 11: Polish, Demo, and Submission

The final week focuses on stability, demo quality, and submission preparation. No new features — only hardening what exists and preparing the presentation.

Milestone	Deliverable	Status
Bug fixing	Full pipeline run without critical failures	To Do
UI polish	Clean, professional coach dashboard on Replit Pro URL	To Do
Demo video	2 to 5 minute demo video per judging requirements	To Do
README	Clear GitHub README with setup instructions and architecture diagram	To Do
Submission	Project submitted via Encode Club platform before June 11	To Do

Finale event	Present at Somnia Agentathon Finale (June 11)	Scheduled
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7. Judging Criteria Alignment

Criterion	How Velo Addresses It
Functionality	Two fully functional agents, end-to-end pipeline from video upload to on-chain receipt, deployed on Replit Pro with a stable public URL.
Agent-First Design	Somnia smart contract autonomously invokes agents — no human step between data submission and on-chain receipt. Agents are registered on Somnia's discovery layer and callable by other on-chain actors.
Innovation and Technical Creativity	First platform combining VLM-based biomechanical reasoning (Marlin-2B) with on-chain verifiable coaching prescriptions. Novel use of Somnia agent primitives for sports intelligence.
Autonomous Performance	Once a video is submitted, no human action is required. The contract triggers Agent 1, Agent 1 triggers Agent 2, Agent 2 writes on-chain. The system runs independently from submission to receipt.

8. Risks and Mitigations

Risk	Likelihood	Mitigation
Marlin-2B inference too slow on HuggingFace free tier during demo	Medium	Run Marlin-2B locally and expose via ngrok for the demo. Zero cost, full control.
Somnia testnet instability or agent API changes	Low-Medium	Attend May 22 workshop, follow Somnia docs closely, build contract interactions with error handling.
MediaPipe accuracy on non-standard camera angles	Medium	Calibrate angle thresholds using self-recorded footage. Document known limitations in README.
Scope creep reducing demo quality	Medium	Strictly enforce the two-agent scope. Ball tracking and live feed are explicitly deferred to post-hackathon.
Replit compute limits under video processing load	Low	Replit Pro provides sufficient compute. Process video asynchronously with a job queue if needed.

Velo — Athletic Intelligence

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